

Уктерган Эйлер: $\int \frac{u^{2n-1}}{1+u^{2k}} du$

$$\frac{u^{2n-1}}{1+u^{2k}} = \frac{u^{2n-1}}{\prod_{j=1}^k (u^2 - 2u \cos \alpha(2j-1) + 1)} = \sum_{n=1}^k \frac{A_n u + B_n}{(u^2 - 2u \cos \alpha(2n-1) + 1)}$$

$\alpha = \frac{\pi}{2k}$

$\alpha = \frac{\pi}{2k}$

$$u^{2k-1} = \sum_{n=1}^k \frac{(A_n u + B_n)(1+u^{2k})}{(u^2 - 2u \cos \alpha(2n-1) + 1)} \Big|_{u=u_n} =$$

$$= \frac{(A_n + B_n)(1+u^{2k})}{u^2 - 2u \cos \alpha(2n-1) + 1} \Rightarrow \text{(no pravuyu sonumada)}$$

$$\Rightarrow \lim_{u \rightarrow u_n} = (A_n u + B_n) \lim_{u \rightarrow u_n} \frac{2k u^{2k-1}}{2u - 2 \cos \alpha(2n-1)} =$$

$$= (A_n u_n + B_n) = u_n^{2k-1} - \cos \alpha(2n-1) =$$

$$= e^{i \alpha(2n-1)} - \cos \alpha(2n-1) = -k A_n e^{i \alpha(2n-1)(1-2k)} - k B_n e^{-i \alpha(2n-1)2k} =$$

$$= \begin{cases} -k A_n \cos(\alpha(2n-1)(1-2k)), \\ -k B_n \cos \alpha(2n-1)2k = 0, \\ -2 A_n \sin(\alpha(2n-1)(1-2k)) + k B_n \sin(\alpha(2n-1)2k) = \\ \quad = \sin \alpha(2n-1); \end{cases}$$

$$\text{Пример 1} \quad \int \frac{x^5 dx}{x^6 - 1} = \left| \quad \frac{1}{6} \int \frac{d(x^6 - 1)}{x^6 - 1} = \right.$$

$$= \frac{1}{6} \ln |x^6 - 1| + C.$$

$$\textcircled{3} \int \frac{x^2 dx}{x^6 - 1} = \frac{1}{3} \int \frac{dx^3}{(x^3)^2 - 1} = \frac{1}{6} \ln \left| \frac{x^3 + 1}{x^3 - 1} \right| =$$

$$\textcircled{2} \int \frac{x^3 dx}{x^6 - 1} = \int \frac{x^3 dx}{(x^3 - 1)(x^3 + 1)} = \frac{1}{2} \int \frac{t dt}{t^3 - 1} = \frac{1}{2} \int \left[\frac{\textcircled{1}}{(t-1)^{\frac{1}{3}}} + \frac{\textcircled{2}t}{t^3 + 1} \right] dt =$$

$$+ \frac{\textcircled{2}}{t+1} \quad \frac{2}{3}$$

$$\textcircled{4} \int \frac{x^4 dx}{x^6 - 1} = \int \frac{x^4}{(x^2 - 1)(x^4 + x^2 + 1)} dx,$$

$$\int \frac{x^4}{(x^3 - 1)(x^3 + 1)} dx = \int \frac{x^4 dx}{(x-1)(x^2+x+1)(x+1)(x^2-x+1)} =$$

$$= \int \left(\frac{\textcircled{1/6}}{x-1} + \frac{Bx+C}{x^2+x+1} + \frac{\textcircled{-1/6}}{x+1} + \frac{Ex+F}{x^2-x+1} \right) dx$$

$$x^2 + x + 1 = 0,$$

$$x = \pm 1 + \sqrt{3}i$$

2

$$x_1^4 = \lim_{x \rightarrow x_1} (Bx_1 + C) \lim_{x \rightarrow x_1} \frac{x^5 - 1}{x^2 + x + 1} = (Bx_1 + C) \lim_{x \rightarrow x_1} \frac{5x^4}{2x + 1} :$$

$$x_1^4 = (Bx_1 + C) \frac{5x_1^4}{2x_1 + 1} ;$$

D/3: система ↑

Метод
осложнений

Метод оспаривающего

$$\begin{aligned} x + 1 &= 2, \\ x &= 1. \end{aligned}$$